

## **Don't Just Prompt—Suggest:**

### **Suggested Plans Boost Planning Prompt Effectiveness in the Field**

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See the [Web Appendix](#) for supplementary materials

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## ABSTRACT

Prompting consumers to plan when and where they will follow through on a goal can close intention-behavior gaps. Such “planning prompts” typically require consumers to form their own plans, and prior research suggests this should enhance follow-through because self-set plans can increase commitment and memorability while simultaneously respecting consumer autonomy. However, we propose that adding a suggested plan—specifying a proposed time, day, and location for action—alongside a planning prompt can further boost follow-through. We theorize that a suggested plan increases the ease of planning by reducing complexity and cognitive effort, which could be more important than bolstering consumer autonomy in the planning process. In a field experiment ( $N = 1,012,143$ ) with a national U.S. pharmacy chain, we find that planning prompts including a suggested plan led to 2.45% more vaccinations than planning prompts alone. In a second field experiment ( $N = 985,145$ ), we show that planning prompts offering a suggested plan outperformed best-practice reminders by 3.26%. Our studies also provide evidence that a planning prompt offering a suggested plan acts as a reference point or anchor, a heuristic that can simplify their decision process; consumers are more likely to get vaccinated both at and near the time of day and day of the week mentioned in the suggestion, compared to customers who only received a planning prompt without a suggested plan.

## INTRODUCTION

Every day, in every one of us, a quiet battle is waged between what we intend to do and what we actually do—a struggle in which our best intentions often need a helping hand. Marketers and policymakers, who frequently seek to promote behaviors that serve consumers' best interests, often play an active role in motivating actions that consumers might not otherwise perform. For example, gyms, healthy lifestyle apps, and fitness programs spend hundreds of millions of dollars to encourage consumers to sign up for and stick with their products (McKinsey & Company 2025). Similarly, providers of educational, healthcare, and financial planning tools work to convert consumers' stated intentions—such as a desire to learn, obtain preventative care, or save for retirement—into concrete actions. Such efforts, however, often yield only limited success (Rogers et al. 2015; Sheeran 2002). Follow-through challenges, which stem from what has been dubbed the “intention-behavior gap” (Sheeran and Webb 2016), have led researchers to study a wide range of strategies for successfully encouraging consumers to act on their intentions.

One approach that has proven particularly effective at encouraging consumers to follow through on good intentions is to use “planning prompts” (Rogers et al. 2015). Such prompts encourage consumers to detail their “implementation intentions”—if-then plans that link a cue, such as a specific time, with a desired action (e.g., if it is Monday at 4pm, then I will go to the pharmacy to receive a vaccination; Gollwitzer 1999). Planning prompts have been shown to encourage desirable behaviors across a range of domains from promoting voter turnout (Nickerson and Rogers 2010) to vaccine take-up (Milkman et al. 2011). But much of the literature on the psychology of planning rests on an implicit assumption: that open-ended

planning prompts—those which invite consumers to generate their own cues in order to construct a plan—are ideal (Sheeran, Listrom, and Gollwitzer 2025). In this paper, we ask: Could planning prompts be even more effective if paired with a suggested plan?

To investigate whether suggested plans enhance a planning prompt's effectiveness, we conducted a large, preregistered field experiment with a national U.S. pharmacy ( $N = 1,012,143$ ) aimed at increasing flu vaccination uptake. In a second field experiment, focused on nudging COVID-19 bivalent booster vaccination ( $N = 985,145$ ), we assessed whether planning prompts paired with a suggested plan could outperform current best practice reminders used to promote vaccination.

Despite the proven cost-effectiveness of vaccines as a means of decreasing disease morbidity, mortality, and healthcare expenses (Nichol et al. 1994), more than half of the eligible U.S. population regularly forgoes the recommended influenza vaccine (CDC 2024) and more than three quarters forgo the recommended yearly COVID booster (CDC 2025). Therefore, our experiments were conducted in a policy-relevant context with significant public health implications. Notably, this is a consequential setting where many other behavioral scientists have also chosen to study follow-through (see for example Chapman et al. 2016 and Dai et al. 2021).

This paper makes four key contributions to the consumer behavior and psychology literatures. First, we advance research on the psychology of planning (Gollwitzer and Sheeran 2025; Rogers et al. 2015). Whereas prior work has typically assumed that inviting consumers to create their own plans is ideal (Sheeran et al. 2025), we explore the possibility that offering a suggested plan alongside a planning prompt may further enhance follow-through by reducing cognitive effort and simplifying planning. Second, we contribute to a growing stream of research on how suggestions influence consumer behavior (Brockner et al. 1984; Goswami and Urmitsky

2016; Moon and VanEpps 2023). We examine how suggestions operate in the underexplored domain of planning and test for the possibility that consumers may anchor and adjust from suggested plans, akin to the way they anchor and adjust from suggested prices and donations (Fraser, Hite, and Sauer 1988; Jung, Perfecto, and Nelson 2016). Third, we contribute to an emerging marketing literature examining the pros and cons of personalization (Shchetkina and Berman 2024; Yoganarasimhan, Barzegary, and Pani 2023). Specifically, we evaluate whether offering a consumer a suggested plan based on group norms or a plan based on that specific consumer's past behavior is more effective, which sheds new light on the value of personalization in an important health setting. Finally, we contribute to the literature on nudging (Johnson et al. 2012) and to marketing practice by evaluating a new, low-cost intervention designed to help marketers bridge the intention-behavior gap.

The remainder of this paper is organized as follows: We begin by (1) developing a theory of why planning prompts offering a suggested plan may be more effective than standard planning prompts and (2) reviewing the relevant literature to motivate this prediction. Next, we present a large, preregistered field experiment (Study 1) that tests this theory. We then report a preregistered second field experiment (Study 2) that establishes the practical relevance of our intervention by comparing its effectiveness against a best-practice benchmark. Finally, we discuss our findings and their implications for theory, for marketers, for consumers, and for future research.

## **THEORETICAL BACKGROUND**

## Planning Prompts

Research on goal-setting and planning shows that consumers who are encouraged to create “implementation intentions”—if-then plans that associate a situational cue (e.g., Monday at 4pm) with a planned, goal-directed response (e.g., getting vaccinated)—are more likely to follow through on their intended actions (Gollwitzer 1999; Gollwitzer and Sheeran 2006; Rogers et al. 2015). For example, in one field experiment, researchers found that mailing employees a flu vaccination reminder that included a prompt to write down the date and time they planned to get vaccinated increased flu inoculation by 4.2 percentage points, compared to mailing employees the same reminder without a prompt to plan (Milkman et al. 2011). Similar “planning prompts” have also encouraged follow-through on voting, colonoscopy uptake, exercise, dieting, and recycling (for a review, see Rogers et al. 2015).

Why are planning prompts effective? First, they make it easier for consumers to remember their intentions at the critical moment. Associating a cue (e.g., Monday 4 pm) with an action (vaccination) makes people more likely to follow through when the cue arises—because the cue serves as a salient reminder, it triggers recall and automatic enactment of the planned action (Gollwitzer 1999; Rogers et al. 2015). Second, merely asking people whether they intend to complete an action makes them reflect on it more than they would otherwise, and such contemplation increases the likelihood that a person will subsequently enact a behavior (Morwitz, Johnson, and Schmittlein 1993). Third, planning prompts increase goal commitment. The act of adopting a plan involves committing to a specific implementation strategy (Gollwitzer 1990), which past research suggests increases effort to follow-through on the plan or implementation intention and, therefore, make completing a goal more likely (Klein, Molloy, and

Brinsfield 2012). Further, making a plan can prevent indefinite procrastination, as committing to a specific time or place for completing an action means that delaying execution would explicitly break a past commitment, which consumers can find aversive (Festinger 1957). Finally, making a plan encourages consumers to think through previously unforeseen obstacles in advance (e.g., the need for childcare or transportation) so these obstacles are more likely to be surmounted if and when they arise: being prompted to proactively unpack when and how consumers will take intended actions can reduce overconfidence in planning (Kruger and Evans 2004).

One common feature of the planning prompts studied in prior research is that they delegate the entire task of making a plan to the consumer (e.g., Milkman et al. 2011, 2013; Nickerson and Rogers 2010). Such delegation implicitly prioritizes maximizing consumer autonomy to facilitate the planning process (Botti, Iyengar, and McGill, 2023; Wertenbroch et al. 2020) and it may enhance the efficacy of planning prompts for several reasons. First, planning prompts are theorized to work, in part, by increasing the ease of recalling intentions when the time has come to act on them (i.e., by increasing memory accessibility; Gollwitzer 1999; Gollwitzer and Sheeran 2006; Orbell, Hodgkins, and Sheeran 1997). Since memories are more accessible when information is self-generated (Slamecka and Graf 1978) and when people engage in problem-solving (Jacoby 1978), it may be beneficial for consumers to make their own plans, without assistance, for planning prompts to boost recall. Furthermore, when participants generate their own plans (rather than acting on plans provided by others) they may be more likely to select cues to trigger action that are personally relevant and salient (e.g., choosing to take medication immediately after finishing their morning coffee), which makes such distinct cues more likely to be remembered (Rogers and Milkman 2016). In addition, providing open-ended prompts caters to the basic human need for autonomy (Ryan and Deci 2017), which may

avoid triggering consumer reactance—a negative response to influence attempts that can occur when people feel their freedom to choose is threatened (Clee and Wicklund 1980). By encouraging consumers to make their own choices, open-ended prompts may also increase consumers' satisfaction with their outcomes (Botti and McGill, 2011). And finally, allowing consumers to form their own plans may increase their commitment to follow through. Since having volition is vital for fostering commitment (Salancik 1977), encouraging consumers to create their own plans makes them less likely to abandon them: unlike suggested or assigned plans, volitional plans are more difficult to discount or dismiss (Hollenbeck and Klein 1987). In sum, in light of the theory that the efficacy of plans is moderated by goal commitment (Gollwitzer and Sheeran 2025), and because self-generated plans are predicted to increase goal commitment for the reasons outlined above, self-generated plans could be expected to improve follow-through. And, in fact, one large meta-analysis of planning interventions found that allowing consumers to select their own cues (to trigger action) while generating plans was correlated with higher follow-through (Sheeran et al. 2025).

### Incorporating Suggestions into Planning

Rather than using a prompt that delegates the formulation of specific plans entirely to the consumer, marketers could offer a suggestion for a plan alongside the planning prompt (e.g., “would getting a vaccine at 4 pm on Monday at 1 Main Street work for you?”). We define “suggestions” as external proposals provided to consumers—such as a suggested time for vaccination or a suggested tip amount. Such suggestions have been shown to promote the

adoption of the suggested option(s) (e.g., a 20% tip) by making it more salient to consumers than the universe of other alternatives (Haggag and Paci 2014).

Suggestions differ from defaults, or pre-selected options that will be automatically enacted, in that suggestions require the consumer's active choice for enactment (Johnson and Goldstein 2013; Chapman et al. 2010). While providing a default has also proved effective for increasing follow-through (e.g., on vaccination; Chapman et al. 2010, 2016), defaults rely on a different mechanism than suggestions: they take advantage of inattention and hassle costs that discourage consumers from overriding a pre-selected option (Thaler, Sunstein and Balz 2013). Suggestions, by contrast, do not rely on these opt-out frictions that guide behavior. Although suggestions may sometimes be presented as a menu of multiple options (Alexander, Boone, and Lynn 2021; Goswami and Urminsky 2016; Haggag and Paci 2014; Moon and VanEpps 2023), we focus specifically on providing a single suggestion—i.e., a suggested plan—within a planning prompt.

Prior empirical research on the effect of offering consumers a single suggestion has largely examined suggestions' impact on consumers' spending decisions and product choices. For instance, researchers have extensively explored the effect of offering a single suggested donation amount on consumers' charitable donations (see, for example, Brockner et al. 1984 and Fraser et al. 1988). They have also examined how suggesting a product affects consumer choices (Fitzsimons and Lehmann 2004; Goodman et al. 2013). On the whole, provided the suggestion does not conflict with a consumer's previously expressed preferences (Fitzsimons and Lehmann 2004; Goodman et al. 2013), these studies show that a suggestion makes consumers more likely to follow through on actions such as donating or making a product purchase.

## Does Including a Suggested Plan Make Planning Prompts More Effective?

We theorize that adding a suggested plan to a planning prompt has the potential to increase consumers' follow-through on their intentions by simplifying the decision-making process. There are two reasons for this conjecture.

A suggested plan may reduce the cognitive effort required to develop, evaluate, and decide on a plan. Consumers often lack well-defined preferences (Bettman, Luce, and Payne 1998) and are more likely to delay decisions when faced with many attractive alternatives (Dhar 1997). Since deferral becomes more likely as the number of options proliferates (Tversky and Shafir 1992), highlighting one viable option may reduce the mental burden of choosing and lower the likelihood of avoiding a decision to plan altogether. Moreover, a salient suggested plan—by virtue of being more easily recalled than other plan options—may be evaluated first, reducing the likelihood that consumers seek out or consider alternative options (Weber and Johnson 2006; Wyer and Srull 1986). As a result, providing a single suggested plan allows consumers to make a straightforward decision without incurring the search costs associated with expanding their choice set (Hauser and Wernerfelt 1990; Shugan 1980).

In addition, a suggested plan may “endow” consumers with a status-quo plan (e.g., Monday at 4 pm for vaccination) and create an anchor or reference point (Tversky and Kahneman 1974). Adjusting from an anchor is a heuristic people often use to reduce decision complexity (Tversky and Kahneman 1974), so if a suggested plan is treated as a reference point, it could be perceived to simplify planning. When people anchor on a suggestion, they are more likely to accept that exact suggestion or to select options adjacent to the suggested amount (as long as the anchor suggested is close to the distribution of consumer choices; Jung, Perfecto, and

Nelson 2016). This means that if consumers treat a suggested plan as an anchor and use simplified decision heuristics (e.g., Bettman et al. 1998), providing a suggested plan featuring a day or time should result in greater follow-through at the precise day and time suggested and at neighboring days and times.

### Operationalizing Suggested Plans

While we theorize that suggested plans can increase the ease of follow-through, the effectiveness of offering a suggested plan likely depends on the quality of the suggestion. We consider two different types of suggested plans: (1) those based on descriptive social norms (e.g., popular day of week/time combinations for vaccination; Cialdini, Kallgren, and Reno 1991), which we call “popular suggested plans” and (2) those personalized based on a participant’s past behavior (e.g., the same date/time combination they selected for their most recent adoption of the recommended action; Arora et al. 2008), which we call “personalized suggested plans.” Past research has shown that suggestions are more effective if they match consumer preferences (Fitzsimons and Lehmann 2004) and that consumers generally respond more positively to personalized stimuli (Ansari and Mela 2003). This suggests that using a consumer’s past choices to infer the best suggestion could be ideal (Johnson and Goldstein 2013). On the other hand, to the extent that consumers’ preferences are unstable over time, such “personalized” suggestions may backfire for consumers whose situation or preferences have shifted: suggestions that are misaligned with a consumer’s current preferences can be a source of annoyance (Goldstein et al. 2008). Our experiments allow us to compare the efficacy of suggesting personalized versus popular plans to consumers.

## Prior Empirical Research

To date, empirical evidence examining how adding a suggested plan to a planning prompt influences follow-through has been limited and inconclusive. One recent field experiment assessing reminders as a means to encourage vaccination (Meeldijk et al. 2025) found that sending reminders suggesting a date, time, and location for a vaccine increased vaccine uptake more than either (1) sending reminders that used entirely different persuasion tactics to encourage vaccination or (2) sending no reminder. However, this study did not compare the effect of offering a suggested plan with the effect of prompting consumers to form their own plan. Other prior research has examined the benefits of providing menus of suggested plans: for example, “volitional help sheets” prompt consumers to make a plan by offering a list of suggested situational cues (e.g., “If I am tempted to smoke at a bar...”) and suggested actions (e.g., “Then I will think about something else”; Armitage 2008). Several studies have compared providing these menu-based suggestions against merely inviting people to self-generate their plans (Armitage 2009; Armitage et al. 2016; Armitage and Arden 2012). However, these studies featured small sample sizes (fewer than 75 participants per condition) and self-reported rather than objective outcomes. Moreover, the measured impact of offering volitional help sheets over open-ended planning prompts has been mixed. And neither prior research stream has compared the effectiveness of suggested plans based on descriptive social norms or past customer behavior.

Prior research has also examined the impact of providing consumers with default vaccination appointments (e.g., “an appointment has been prescheduled for you at the following date, time, and location”; Chapman et al. 2010, 2016). Although default appointments share some similarities with suggested plans (e.g., both provide a reference point for when and where

to act), defaults take advantage of consumer inertia, inattention, or hassle costs (Thaler, Sunstein, and Balz 2013) because they do not require an action to set an appointment. By contrast, suggested plans require active attention to opt-in to an appointment and thus cannot capitalize on inertia or inattention and do little to reduce hassle costs. While defaults are often not practical to implement (our field partners in Studies 1 and 2, for example, would not consider allocating patients with default appointments because a 71% no-show rate as reported in Chapman et al. 2016 would have been too costly to manage), their efficacy offers some evidence that simplifying the planning process via the provision of a suggested plan could be valuable.

Our research extends prior research by investigating whether offering a single suggested plan alongside a planning prompt improves follow-through more than a standard planning prompt alone in a consequential setting. Across two preregistered large-scale natural field experiments that employ objective outcome measures to maximize policy relevance and contextual realism (Lieberman, Morales, and Amir 2019; Ratner et al. 2008), we also provide evidence that suggested plans act as a reference point or an anchor, shifting the way in which people follow-through on their intentions when they are encouraged to plan.

## OVERVIEW OF EXPERIMENTS

We investigate whether a planning prompt offering a suggested plan can enhance follow-through on vaccine uptake in two preregistered field experiments ( $N = 1,997,288$ ), each conducted in partnership with a large U.S. pharmacy chain.

Study 1 ( $N = 1,012,143$ ) is a field experiment that directly tests our theory by comparing the effectiveness of two types of planning prompts: (1) a standard, open-ended, planning prompt

vs. (2) a planning prompt that offers a suggested plan (i.e., a suggested time of day, day of the week, and location for vaccination). We also examine what proportion of consumers precisely follow our suggested vaccination plans as well as what proportion anchor and adjust slightly from them. Finally, Study 1 allows us to compare the benefits of suggested plans derived from descriptive social norms versus past behavior.

Study 2 tests the policy-relevance of deploying planning prompts offering a suggested plan. In a second preregistered field experiment ( $N = 985,145$ ), we evaluate whether planning prompts offering a suggested plan outperform current best practice reminders, which include neither a planning prompt nor a suggested plan. Additionally, we seek to replicate results from Study 1 regarding consumers' tendency to anchor and adjust from suggested plans. Though made possible by the infrastructure of a larger megastudy on vaccine adherence, Study 2 was executed as a self-contained experiment, with a separate theoretical motivation and corresponding preregistration developed by the authors of this paper. In contrast to the current investigation, the aggregate results of the megastudy have been reported alongside a half-dozen other interventions in meta-analytic form, primarily emphasizing the lack of benefit provided by offering consumers free Lyft rides to and from vaccination appointments (Milkman et al., 2024).

Both studies 1 and 2 were approved by the IRB at [name removed to preserve blind review], which granted waivers of consent due to the study's minimal risk to participants and the infeasibility of following a standard research consent process in this field setting. The partner pharmacy for each study provided all data in a de-identified format in accordance with the Safe Harbor method pursuant to 45 C.F.R. § 164.514(b)(2).

## **STUDY 1: ARE PLANNING PROMPTS OFFERING A SUGGESTED PLAN MORE EFFECTIVE THAN STANDARD PLANNING PROMPTS?**

In the fall of 2023, we partnered with a national U.S. pharmacy chain to send text messages to more than 1 million customers, encouraging them to get vaccinated for the upcoming influenza season. Our study design and analysis plan were preregistered on the Open Science Framework (initial preregistration: <https://bit.ly/4b6TmQ3>; updated preregistration: <https://bit.ly/45wHJkr>).<sup>1</sup> The cleaned and de-identified dataset, analysis code, log files, and table outputs for Study 1 are all posted on OSF (<https://bit.ly/3KmXZhl>).

### **Participants**

Customers were eligible to be included in our experiment if they were (1) 18 years or older, (2) had consented to receiving text message communications from the pharmacy, (3) had received at least one vaccination, of any type, in the past two years according to pharmacy records, and (4) had not already received a flu vaccine at the pharmacy during the fall 2023 flu season. Eligibility was also constrained by our pharmacy partner's standard messaging rules,

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<sup>1</sup> Our initial preregistration (<https://bit.ly/4b6TmQ3>) was submitted on September 6<sup>th</sup> before the launch of our experiment. After our partner organization identified a technical error in message delivery during the launch of the experiment, we submitted a second updated preregistration (<https://bit.ly/45wHJkr>) before the experiment concluded (on Monday, October 16<sup>th</sup>, 2023) and before any data was shared (on Friday, October 20<sup>th</sup>, 2023). For additional details on our handling of the technical error, please see the "Procedures" section and Web Appendix D.

including a requirement that customers had not recently received any other vaccination text messages in the days immediately prior to our study's launch. Based on these combined eligibility criteria, a total of 1,012,143 customers were selected for inclusion in our experiment.<sup>2</sup>

As shown in Web Appendix table C1 (which provides detailed summary statistics on our study population), 59.3% of the customers in our experimental sample were female and 40.7% were male. The average age of these customers was 51.65 (SD = 17.95), and 38.9% had received a flu vaccine at our partner pharmacy in the prior 2022-2023 flu season.

## Procedures

As shown in figure 1, eligible customers were randomly assigned to one of two primary experimental conditions, which determined whether they received: (1) a text message prompt to plan when and where they would get vaccinated (*standard planning prompt* condition) or (2) a text message prompt to plan when and where they would get vaccinated alongside a suggested plan (*planning prompt offering a suggested plan* condition). Customers in our *planning prompt offering a suggested plan* condition were further randomized into two different sub-conditions. For 337,306 customers, we suggested a plan based on the day of the week and time of day when

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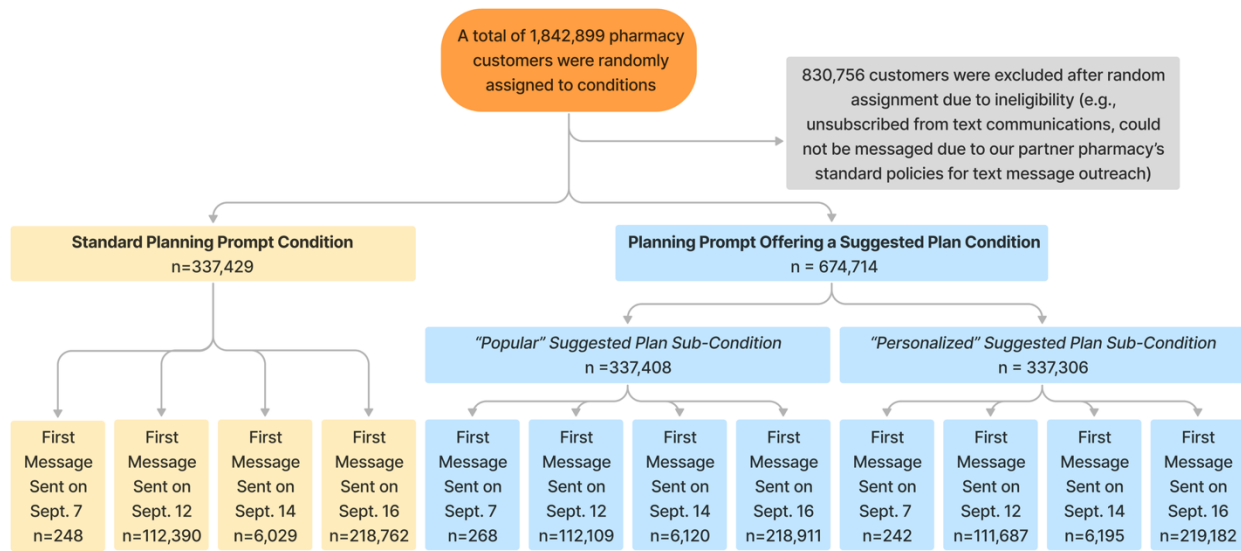
<sup>2</sup> Our preregistration stated that we would seek to include 1.8 million customers in this study and that our sample size would match the maximum number of customers our pharmacy partner was able to message given business priorities and constraints. We randomized 1,842,899 consumers on August 17, 2023; our pharmacy partner was able to message 1,012,143 customers who met our eligibility criteria under our pharmacy partner's standard outreach rules.

they received their last vaccination at our pharmacy partner (a “personalized” suggestion). For 337,408 customers, we instead suggested a randomly selected day-of-the-week and time-of-day combination drawn from a list of the 10 most popular times when a different set of customers at a national U.S. pharmacy chain had chosen to get vaccinated in the prior year (a “popular” suggestion).<sup>3</sup> There was little overlap between the day of the week and time of day offered in “popular” and “personalized” suggested plans; only 11.2% of personalized suggestions matched any of the ten most popular day-of-the-week and time-of-day combinations. For both the “popular” and “personalized” sub-conditions of the *planning prompt offering a suggested plan* condition, the suggested vaccination plan recommended receiving a vaccination at the pharmacy location where the customer had received their most recent vaccination at our partner pharmacy.

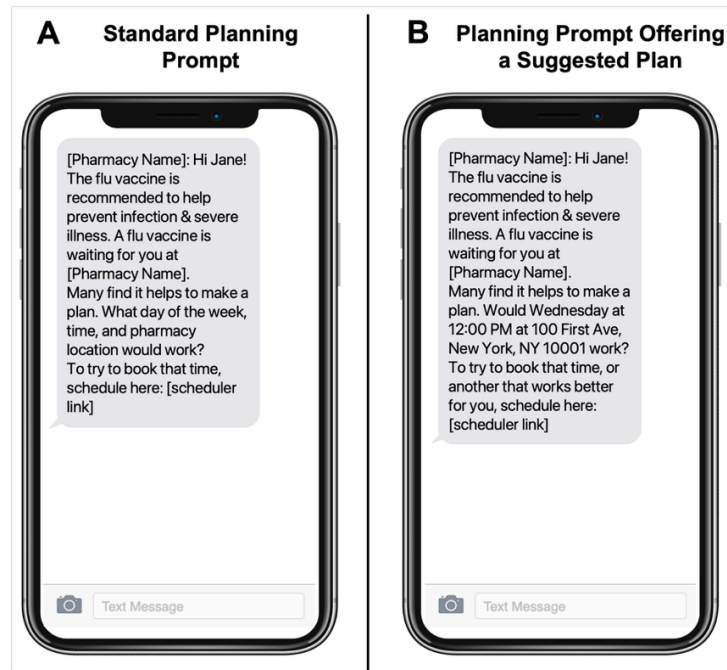
All customers in this study received a text message stating that a flu vaccine was “recommended” and “waiting for you” at the pharmacy and were then prompted to make a plan for getting their vaccine. As shown in figure 2, the *standard planning prompt* message asked customers: “What day of the week, time, and pharmacy location would work?” In contrast, the *planning prompt offering a suggested plan* message asked customers: “Would [day of the week] at [time of day] at [pharmacy address] work?” For example, it might read: “Would Friday at 03:00 PM at 1 Main Street, New York, NY 10001 work?”

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<sup>3</sup> The ten most popular day-of the-week/time-of-day combinations were (in order of popularity): (1) Friday at 3 PM, (2) Friday at 11 AM, (3) Wednesday at 3 PM, (4) Monday at 11 AM, (5) Wednesday at 11 AM, (6) Monday at 3 PM, (7) Friday at 4 PM, (8) Friday at 12 PM, (9) Wednesday at 12 PM, or (10) Monday at 12 PM. See Appendix A for details on how these times were determined.

**FIGURE 1****FLOW DIAGRAM FOR IMPLEMENTATION OF STUDY 1**

NOTE.—This CONSORT (Consolidated Standards of Reporting Trials; Moher, Schulz, and Altman 2001) diagram shows how our experimental design was implemented by our pharmacy partner. Our design randomly assigned customers to two different primary study conditions (*standard planning prompt* condition and *planning prompt offering a suggested plan* condition). Customers in the *planning prompt offering a suggested plan* condition were then further randomly subdivided into two different sub-conditions offering either a “popular” or “personalized” suggested plan. Due to a technical error, some customers who received their first message on September 12 also received a duplicate message on September 14: the counts of affected customers by group were: (1) *standard planning prompt* condition:  $n = 52,466$ ; (2) “popular” suggested plan sub-condition:  $n = 51,998$ ; (3) “personalized” suggested plan sub-condition:  $n = 51,916$ .

**FIGURE 2****TEXT MESSAGES SENT TO CUSTOMERS BY CONDITION IN STUDY 1**

NOTE.—In panel B, the suggested day of the week and time of day was either (1) randomly selected from the ten most popular day of the week and time combinations for vaccination among a prior set of customers at a national U.S. pharmacy chain or (2) matched to the day of the week and time of the customer’s most recent vaccination. In both panels, the recommended pharmacy location always matched the location where the customer received their most recent vaccination with our pharmacy partner.

Customers were randomly assigned to receive text messages on multiple dates in September 2023, beginning with a small pilot and followed by larger rollouts. Due to technical errors, not all customers received messages as intended. As shown in the CONSORT diagram in figure 1, our partner delivered messages to (1) 758 customers on September 7th (as intended), (2) 336,186 customers on September 12th (as intended), (3) 18,344 customers on September 14th

(who were intended to receive their message on September 12th), and (4) 656,855 customers on September 16th (some of whom were intended to receive their message on September 14th and some of whom were intended to receive their message on September 16th). The technical errors in implementation not only included the imperfect deployment dates of the initial text messages described above, but also produced a duplicate text message sent on September 14th to 156,380 customers who had already received a message on September 12th. For additional details on these errors, see Web Appendix D. To address these errors, we conducted balance checks assessing differences in observable characteristics for each group of customers by reminder send date and by number of text messages received, and these checks do not reveal significant overall imbalances on observables (see Web Appendix table D1). In our analyses, we combined all customers from the experiment into a single group, regardless of whether they received one or two text messages from the pharmacy. We then estimated a single treatment effect on an intent-to-treat basis—both with and without controlling for the customers who received an additional reminder message. We also ran robustness tests confirming that our main preregistered findings remain consistent even after excluding all customers who were originally randomized to receive a message on September 12th (and could have been affected by the technical errors).

### Dependent Variables and Analysis Strategy

As shown in table 1, we defined six categories of dependent variables. We analyzed whether random assignment to the *planning prompt offering a suggested plan* condition in our experiment affected each dependent variable as follows:

**TABLE 1****DEFINITION OF DEPENDENT VARIABLES IN STUDY 1**

| <b>Dependent Variable</b>                                                 | <b>Definition of Dependent Variable</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Vaccination Appointments Scheduled</i>                                 | We preregistered analyzing clicks on the vaccine scheduler link provided in our text messages by condition; however, because we could not obtain this measure, we instead analyze the closest available variable we received in our dataset—a “vaccine appointment rate”—defined as number of appointments booked via the provided vaccine scheduler link divided by the total number of customers texted in that condition (meaning, we assumed that each customer booked at most one appointment).                                                                                                                                                                                                                                                                                      |
| <i>Vaccinations Received</i>                                              | Our preregistered primary dependent variable was whether a participant received an influenza vaccination at our pharmacy partner within 30 days of the send date of their initial reminder. Two secondary dependent variables measuring vaccination receipt were also preregistered: (1) receipt of an influenza vaccination within 60 days and (2) receipt within 90 days of the send date of a customer’s initial reminder message.                                                                                                                                                                                                                                                                                                                                                     |
| <i>Vaccinations Received by Following Our Suggested Personalized Plan</i> | For personalized suggested plans, each customer was offered a suggested plan that matched the day of the week, time of day, and location of their most recent prior vaccination. Accordingly, we preregistered three separate binary dependent variables for exploratory analyses: whether customer received an influenza vaccination at our pharmacy partner within 30 days of the send date of their initial reminder (1) on the same day of the week as their last vaccination, or (2) within 15 minutes of the same time of day as their last vaccination, or (3) at the same pharmacy location as their last vaccination.                                                                                                                                                            |
| <i>Vaccinations Received by Following Our Suggested Popular Plan</i>      | For popular suggested plans, each customer was offered a plan to get vaccinated on a popular day-of-the-week (either Monday, Wednesday, or Friday; for additional details see the “Procedures” section or Web Appendix A) and a popular time of day (i.e., either 11AM, 12PM, 3PM or 4PM), along with the location of their most recent prior vaccination. We preregistered three separate binary dependent variables for exploratory analyses: whether customers received an influenza vaccination at our pharmacy partner within 30 days of the send date of their initial reminder (1) on a popular day of week (Monday, Wednesday or Friday), or (2) within 15 minutes of a popular time of day (11AM, 12PM, 3PM, or 4PM), or (3) at the pharmacy location of their last vaccination. |

|                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><i>Vaccinations<br/>Received by<br/>Deviating Slightly<br/>from Our Suggested<br/>Personalized Plan</i></p> | <p>In analyses that were not pre-registered, we analyze whether customers received an influenza vaccination at our pharmacy partner within 30 days of the send date of their initial reminder in a time period close to the suggested plan. We defined four separate binary variables to measure “closeness”: receiving the vaccine (1) on the day before or after the day-of-the-week of their last vaccination; (2) within 15–30 minutes, or (3) within 15–60 minutes, or (4) within 15–120 minutes prior to or after the time-of-day of their last vaccination.</p>                                                            |
| <p><i>Vaccinations<br/>Received by<br/>Deviating Slightly<br/>from Our Suggested<br/>Popular Plan</i></p>      | <p>In analyses that were not pre-registered, we created a set of four separate binary dependent variables for each of the ten popular day-of-the-week/time-of-day suggested plans. For each of the ten popular day-of-the-week/time-of-day combinations (e.g., Monday at 11AM), we analyze whether customers received a vaccination for (1) the day before or after the day-of-the-week specified in that popular suggested plan (e.g., Monday); (2) within 15–30 minutes, or (3) within 15–60 minutes, or (4) within 15–120 minutes prior to or after the time-of-day specified in that popular suggested plan (e.g., 11AM).</p> |

*Vaccination Appointments Scheduled.* We conducted pre-registered two-sample proportion tests to assess whether there were significant differences in the rate at which total vaccination appointments were booked (using the link provided in the text message reminders) as a percentage of the total customers texted across conditions.

*Vaccinations Received.* We preregistered an ordinary least squares (OLS) regression with control variables to maximize the statistical power of our primary confirmatory analyses, as is common in past behavioral science research on vaccination (see Milkman et al. 2021, 2022, and 2024; Dai et al. 2021). Specifically, we fitted an OLS regression with HC1 robust standard errors to predict whether a given customer received an influenza vaccine within 30 days of receiving their text message for our primary analysis. The primary predictor variable in our regression was an indicator for assignment to the *planning prompt offering a suggested plan* condition. Our regression included controls for the customer’s age, an indicator for whether the customer was  $\geq$

50 years old, an indicator for whether the customer was male, an indicator for whether the customer had received their flu vaccination at our pharmacy partner in the 2022-2023 flu season, and indicators for the customer's insurance type (Medicare, Medicaid, other insurance, with an indicator for commercial insurance omitted), all as of August 2023. Following our preregistration, we addressed missing insurance status (which affected data for 0.05% of customers) by (1) replacing missing values in the insurance status indicators (Medicare, Medicaid, other insurance) with the mean among non-missing observations for each variable, and (2) including an additional indicator for missing insurance.

To test whether the effectiveness of the “personalized” and “popular” suggested plans differed, we used the same regression specification but replaced the primary predictor variable with two separate indicator variables: (1) one for assignment to the *personalized suggested plan* sub-condition and (2) one for assignment to the *popular suggested plan* sub-condition (omitting the indicator for the *standard planning prompt* condition). We then conducted a Wald test of the null hypothesis that the coefficients on the *personalized suggested plan* and *popular suggested plan* indicators were equal.

*Vaccinations Received by Following Our Personalized Suggested Plan.* We conducted pre-registered two-sample proportions tests to assess whether customers who were randomly assigned to the *personalized suggested plan* sub-condition were more likely than those assigned to the *standard planning prompt* condition to get vaccinated (1) at the same time of day, (2) on the same day of the week, or (3) at the same location as their most recent prior vaccination.

*Vaccinations Received by Following Our Popular Suggested Plan.* We conducted pre-registered two-sample proportions tests to assess whether customers who were randomly assigned to the *popular suggested plan* sub-condition were likely than those assigned to the *standard planning prompt* condition to get vaccinated (1) on a popular day of week (Monday, Wednesday or Friday), (2) within 15 minutes of a popular time of day (11AM, 12PM, 3PM, or 4PM), or (3) at the pharmacy location of their last vaccination.

*Vaccinations Received by Deviating Slightly from Our Personalized Suggested Plan.* We followed the same analytic approach described in the “Vaccinations Received by Following Our Personalized Suggested Plan” section above. However, we replaced the outcome variables analyzed to focus not on whether customers were vaccinated on *exactly* the same day of the week/time of day/location where and when they received their last vaccine, but instead to assess whether they were vaccinated close to the time of their last vaccination. Specifically, were they vaccinated one day before or after the day of the week of their last vaccination? And separately, were they vaccinated (1) within 15-30 minutes, or (2) within 15-60 minutes, or (3) within 15-120 minutes of the time of their last vaccination?

*Vaccinations Received by Deviating Slightly from Our Popular Suggested Plan.* For popular suggested plans, calculating the proportion of customers who were vaccinated near, but not on, any of the popular days of the week or times of day would cover almost all of the available time windows for vaccination, making it an uninformative measure. Therefore, we employed a different approach.

To test if customers deviated slightly from our suggested day of the week, we performed a separate analysis for each of the ten popular day-of-the-week/time-of-day combinations (e.g., Monday at 11 a.m., Friday at 3 p.m., etc.) separately and then combined them using a meta-analysis. For each of the popular day-of-the-week/time-of-day combinations, we calculated the difference in the likelihood of being vaccinated on the day before or after the focal popular day-of-the-week between customers in two conditions: (1) the subgroup of the *popular suggested plan* sub-condition which offered that specific focal popular day-of-the-week/time-of-day combination being analyzed and (2) the *standard planning prompt* condition. Then, we conducted a meta-analysis of all treatment effects described, using the *meta* package in R, aggregating the ten estimated treatment effects to calculate an average treatment effect.

To test if customers deviated slightly from the suggested time of day, we used the same meta-analytic approach, but replaced the likelihood of receiving a vaccination the day before or after the popular day-of-the-week with the likelihood of receiving a vaccination (1) within 15-30 minutes, (2) within 15-60 minutes, or (3) within 15-120 minutes of the time-of-day of each of the ten popular day-of-the-week/time-of-day combinations.

Our statistical procedure calculates the standard error of the meta-analytic average effect size under the assumption that the ten suggestion-specific estimates are uncorrelated. See Web Appendix G for full details on this meta-analytic approach, including what assumptions we make on the heterogeneity of effect sizes, which method we use to pool variance, what estimators we use, and what adjustments we make to ensure our treatment estimates are robust.

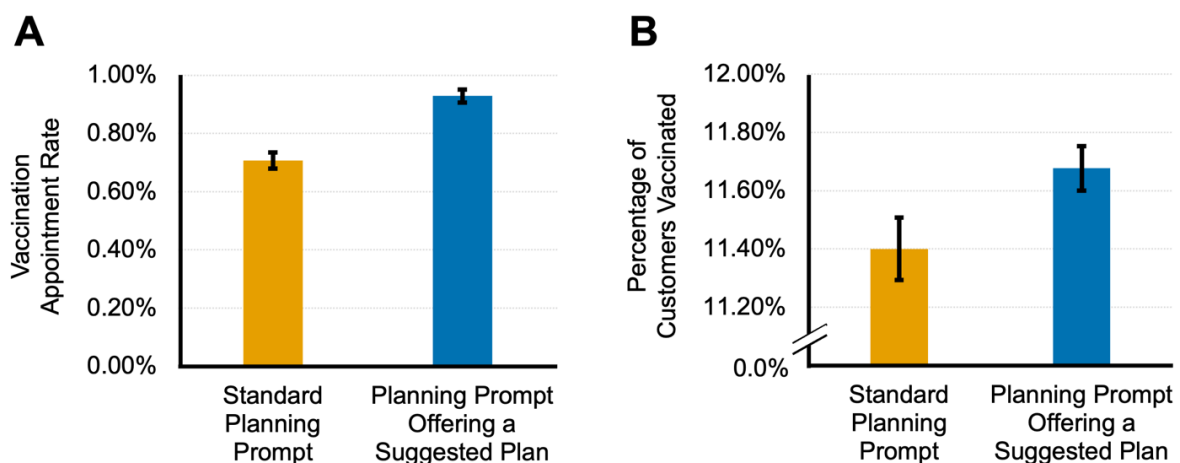
## Results

*Balance Checks.* We confirmed that random assignment was successful, as customer characteristics were balanced on observables across all study conditions (see Web Appendix table C1). We also find no differences in customer covariates across subgroups that were affected by the technical error in our study's deployment (see Web Appendix D).

*Vaccination Appointments Scheduled.* First, we tested whether adding a suggested plan increased the proportion of consumers who completed the planning process (by booking vaccination appointments). As shown in figure 3, panel A, assignment to the *planning prompt offering a suggested plan* condition increased the proportion of customers who booked an appointment using provided text message links from 0.71% to 0.93% (an increase of 0.22 percentage points or 31%;  $\chi^2(1) = 130.15$ ,  $p < 0.001$ ). There was no difference in vaccination appointments booked between the *popular suggested plan* (0.92%) and *personalized suggested plan* (0.93%) sub-conditions ( $\chi^2(1) = 0.13$ ,  $p = 0.723$ ).

**FIGURE 3**

VACCINATION APPOINTMENT RATE (PANEL A) AND  
VACCINATION RATE (PANEL B) BY CONDITION IN STUDY 1



NOTE.—Error bars indicate 95% confidence intervals (CIs) calculated using the normal approximation to the binomial distribution. Panel A shows the vaccination appointment rate, using the scheduling link provided in our text intervention, within 30 days of the study's first text message. Panel B shows the percentage of customers vaccinated at our pharmacy partner within 30 days of their first text message.

*Vaccinations Received.* Next, we tested our hypothesis that adding a suggested plan to a planning prompt increased follow-through on vaccinations. As shown in table 2, model 1, which presents our primary preregistered regression analysis, assignment to the *planning prompt offering a suggested plan* condition produced a regression-estimated 0.28 percentage point increase in customers' vaccination rate to 11.68% (a 2.45% increase from 11.40%;  $\beta = 0.0028$ ,  $t(1,012,133) = 4.32$ ,  $p\text{-value} < 0.001$ ).

As shown in table 2, model 2, our results remain meaningfully unchanged when using the alternative model specified in our second preregistration, which includes an additional control variable: an indicator for whether a customer (accidentally) received two study text messages reminding them to get vaccinated. Our results are also robust to additional preregistered adjustments, including using 60-day and 90-day follow-up windows (our secondary, preregistered dependent variables; see Web Appendix table E1), omitting all control variables from our regression (see Web Appendix table F1), and fitting a logistic regression instead of an OLS regression to our data (see Web Appendix table F2).

As reported in table 2, model 3, the regression-estimated effects of supplementing a planning prompt with a personalized suggested plan (0.25%) versus a popular suggested plan (0.31%) were not statistically significantly different from each other ( $p = 0.418$ ). For further

robustness checks regarding the impact of supplementing planning prompts with a personalized versus popular suggested plan, see table 2, model 4 and Web Appendix F.

**TABLE 2**  
IMPACT OF ADDING A SUGGESTED PLAN TO A PLANNING PROMPT ON FLU  
VACCINATION

|                                                        | <b>Model 1</b>         | <b>Model 2</b>         | <b>Model 3</b>         | <b>Model 4</b>         |
|--------------------------------------------------------|------------------------|------------------------|------------------------|------------------------|
| Planning Prompt Offering a Suggested Plan              | 0.0028***<br>(0.0006)  | 0.0028***<br>(0.0006)  |                        |                        |
| Planning Prompt Offering a Popular Suggested Plan      |                        |                        | 0.0025***<br>(0.0007)  | 0.0025***<br>(0.0007)  |
| Planning Prompt Offering a Personalized Suggested Plan |                        |                        | 0.0031***<br>(0.0007)  | 0.0031***<br>(0.0007)  |
| Age                                                    | 0.0016***<br>(0.0000)  | 0.0016***<br>(0.0000)  | 0.0016***<br>(0.0000)  | 0.0016***<br>(0.0000)  |
| Age $\geq 50$                                          | 0.0063***<br>(0.0011)  | 0.0063***<br>(0.0011)  | 0.0063***<br>(0.0011)  | 0.0063***<br>(0.0011)  |
| Male                                                   | -0.0002<br>(0.0006)    | -0.0002<br>(0.0006)    | -0.0002<br>(0.0006)    | -0.0002<br>(0.0006)    |
| Medicaid Insurance                                     | -0.0166***<br>(0.0009) | -0.0165***<br>(0.0009) | -0.0166***<br>(0.0009) | -0.0165***<br>(0.0009) |
| Medicare Insurance                                     | 0.0340***<br>(0.0011)  | 0.0340***<br>(0.0011)  | 0.0340***<br>(0.0011)  | 0.0340***<br>(0.0011)  |
| Other Insurance                                        | -0.0151***<br>(0.0010) | -0.0151***<br>(0.0010) | -0.0151***<br>(0.0010) | -0.0151***<br>(0.0010) |
| Missing Insurance                                      | -0.0340***<br>(0.0062) | -0.0340***<br>(0.0062) | -0.0340***<br>(0.0062) | -0.0340***<br>(0.0062) |
| Vaccination in Prior Flu Season                        | 0.1487***<br>(0.0007)  | 0.1487***<br>(0.0007)  | 0.1487***<br>(0.0007)  | 0.1487***<br>(0.0007)  |
| Customer Received Two Reminders                        |                        | -0.0045***<br>(0.0008) |                        | -0.0045***<br>(0.0008) |
| Number of Observations                                 | 1,012,143              | 1,012,143              | 1,012,143              | 1,012,143              |
| Adjusted R-Squared                                     | 0.08                   | 0.08                   | 0.08                   | 0.08                   |

\*\*\* $p < .001$ , \*\* $p < .01$ , \* $p < .05$

NOTE.—This table reports the results of four ordinary least squares (OLS) regressions with HC1 robust standard errors. The dependent variable in all models is a binary indicator for whether a customer received an influenza vaccination within 30 days of first receiving an experimental text message. Models 1 and 3 include all pre-registered control variables in our primary regression specification (see the “Dependent Variables and Analysis Strategy” section). Models 2 and 4 include an additional preregistered control variable: an indicator for whether a customer was accidentally sent two reminder text messages (instead of one text message as planned).

*Vaccinations Received by Following Our Suggested Plan.* If suggested plans act as a reference point, consumers should be more likely to receive vaccinations on the exact day of the week, time of day, and location provided by our suggested plans.

As shown in table 3, customers who received a *personalized* suggestion for when to get vaccinated were 0.22 percentage points more likely to receive a vaccination on the same day of the week as their last vaccine (the day recommended in the *personalized suggested plan* sub-condition) than were customers in the *standard planning prompt* condition (a 9.82% increase;  $\chi^2(1) = 35.07$ ,  $p < 0.001$ ). Those who received a *personalized* suggestion for when to get vaccinated were also 0.08 percentage points more likely to receive a vaccination at the same time of day as their last vaccination (the time recommended to them) than were customers in the *standard planning prompt* condition (a 8.88% increase;  $\chi^2(1) = 12.07$ ,  $p < 0.001$ ).

Turning to customers who received a *popular* suggestion for when to get vaccinated (instead of a personalized suggestion), we find they were 0.25 percentage points more likely to receive a vaccination on any popular day of the week than customers in the *standard planning*

*prompt* condition (a 4.23% increase;  $\chi^2(1) = 18.21$ ,  $p < 0.001$ ), and they were 0.11 percentage points more likely to receive a vaccination at any popular time than customers in the *standard planning prompt* condition (a 3.49% increase;  $\chi^2(1) = 6.45$ ,  $p < 0.011$ ).

Because all customers who received a planning prompt offering a suggested plan were encouraged to visit the same pharmacy where they had received their last vaccine regardless of whether they were in the *personalized* or *popular suggested plan* sub-conditions, we pool these sub-conditions to show that these customers were 0.30 percentage points more likely than those in the *standard planning prompt* condition to receive a vaccination at the same pharmacy location where they received their last vaccination (a 4.56% increase;  $\chi^2(1) = 31.59$ ,  $p < 0.001$ ).

*Vaccinations Received by Deviating Slightly from Our Suggested Plan.* If anchoring and adjustment from suggested plans were present, consumers who received a suggested plan should have been more likely to receive vaccinations near, but not exactly on, either the same day of the week and time of day (1) as their last vaccination (in the *personalized suggested plan* sub-condition) or (2) as one of the ten popular day/time combinations suggested in the *popular suggested plan* sub-condition.

Assignment to the *personalized suggested plan* sub-condition—instead of the *standard planning prompt* condition—increased the percentage of customers vaccinated one day before or after the day of the week of their last vaccination by 0.14 percentage points (a 4.27% increase;  $\chi^2(1) = 10.23$ ;  $p\text{-value} = 0.001$ ). Assignment to the *personalized suggested plan* sub-condition also marginally increased the percentage of customers vaccinated within 15-30 minutes before and after the time of their last vaccination (a 0.04 percentage point or 5.11% increase;  $\chi^2(1) = 3.67$ ;  $p\text{-value} = 0.055$ ) and significantly increased the percentage of customers vaccinated within

TABLE 3

TWO-SAMPLE PROPORTION TESTS COMPARING WHEN AND WHERE CUSTOMERS RECEIVED THEIR  
VACCINATIONS ACROSS CONDITIONS

|                                                                                                   | Proportion of Customers Vaccinated at a Given Time, on a Given Day, or at a Given Pharmacy Location |                                             |                                                  |                                             | P-Values from Pairwise Comparisons Using a Two-Sample Proportions Test |                   |                   |
|---------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------------------------|--------------------------------------------------|---------------------------------------------|------------------------------------------------------------------------|-------------------|-------------------|
|                                                                                                   | <i>Planning Prompt Offering a Suggested Plan Condition</i>                                          |                                             |                                                  |                                             |                                                                        |                   |                   |
|                                                                                                   | <i>Standard Planning Prompt Condition</i>                                                           | <i>Suggested Plan Sub-Conditions Pooled</i> | <i>Personalized Suggested Plan Sub-Condition</i> | <i>Popular Suggested Plan Sub-Condition</i> | (1)<br>vs.<br>(3)                                                      | (1)<br>vs.<br>(4) | (3)<br>vs.<br>(4) |
|                                                                                                   | (1)                                                                                                 | (2)                                         | (3)                                              | (4)                                         |                                                                        |                   |                   |
| Vaccinated at Time Suggested in <i>Personalized Suggested Plan Sub-Condition</i>                  |                                                                                                     |                                             |                                                  |                                             |                                                                        |                   |                   |
| On Same Day of the Week as Last Vaccination                                                       | 2.21%                                                                                               | 2.35%                                       | 2.43%                                            | 2.27%                                       | 0.000                                                                  | 0.078             | 0.000             |
| At Same Time of Day as Last Vaccination                                                           | 0.94%                                                                                               | 0.99%                                       | 1.02%                                            | 0.95%                                       | 0.001                                                                  | 0.528             | 0.004             |
| On Same Day of the Week and at Same Time of Day as Last Vaccination                               | 0.22%                                                                                               | 0.25%                                       | 0.27%                                            | 0.23%                                       | 0.000                                                                  | 0.503             | 0.000             |
| Vaccinated at Time Suggested in <i>Popular Suggested Plan Sub-Condition</i>                       |                                                                                                     |                                             |                                                  |                                             |                                                                        |                   |                   |
| On a Popular Day of the Week (Mon, Wed, or Fri)                                                   | 5.80%                                                                                               | 5.98%                                       | 5.92%                                            | 6.04%                                       | 0.039                                                                  | 0.000             | 0.027             |
| At a Popular Time of Day (11am, 12pm, 3pm, or 4pm)                                                | 3.10%                                                                                               | 3.21%                                       | 3.21%                                            | 3.21%                                       | 0.012                                                                  | 0.011             | 0.979             |
| On a Popular Day of the Week and Time of Day (e.g., Mon at 11am)                                  | 1.25%                                                                                               | 1.31%                                       | 1.28%                                            | 1.33%                                       | 0.250                                                                  | 0.002             | 0.058             |
| Vaccinated at Location Suggested in <i>Popular and Personalized Suggested Plan Sub-Conditions</i> |                                                                                                     |                                             |                                                  |                                             |                                                                        |                   |                   |
| At Same Location as Last Vaccination                                                              | 6.50%                                                                                               | 6.80%                                       | 6.81%                                            | 6.79%                                       | 0.000                                                                  | 0.000             | 0.667             |
| Number of Observations                                                                            | 337,429                                                                                             | 674,714                                     | 337,306                                          | 337,408                                     |                                                                        |                   |                   |

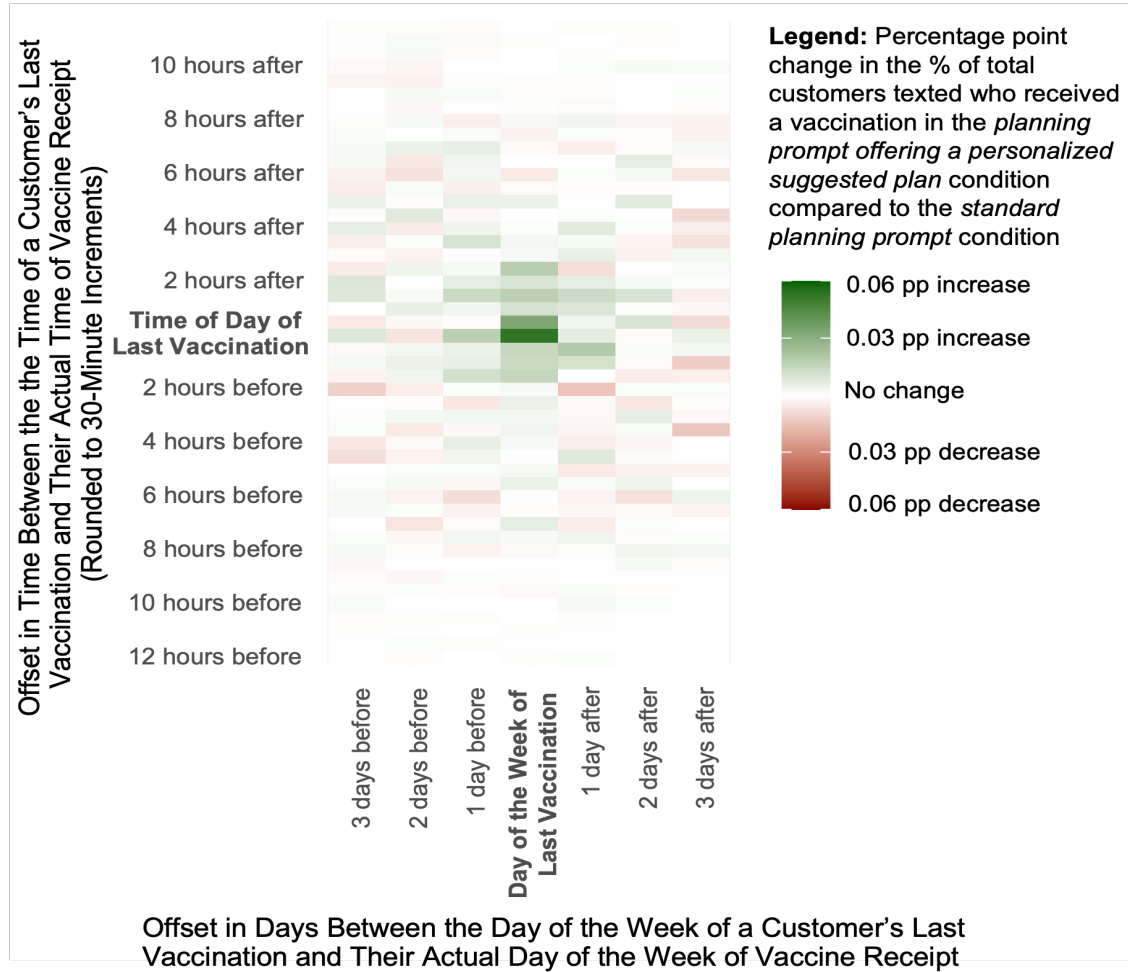
NOTE.— We define customers as vaccinated at the “Same Time of Day as Last Vaccination” if their vaccination occurred within 15 minutes of their prior vaccination time, or at a “Popular Time of Day” if it occurred within 15 minutes of a popular time. Vaccinated “on a Popular Day of the Week and Time of Day” corresponds to receipt of a vaccination within 15 minutes of one of the ten popular day-of-the-week and time-of-day combinations (see footnote 3 in the “Procedures” section for the full list).

15-60 minutes or 15-120 minutes before and after the time of their last vaccination (15-60 minutes before and after: a 0.12 percentage point or 5.13% increase;  $\chi^2(1) = 10.22$ ; p-value = 0.001; 15-120 minutes before and after: a 0.22 percentage point or 4.96% increase;  $\chi^2(1) = 18.53$ ; p-value < 0.001). See figure 4 for a visual representation of these effects, illustrating the shift in vaccination timing for consumers who received a personalized suggested plan.

Turning away from the personalized suggested plans to the popular suggested plans, we see somewhat similar results. Assignment to the *popular suggested plan* sub-condition—instead of the *standard planning prompt* condition—also made it more likely for consumers to receive a vaccination around the time of day of each of the ten popular times, but not around the day of the week of each of the ten popular times (i.e., on average, there was no increase in vaccine uptake on the day before or after each of the ten popular times). Specifically, assignment to the *popular suggested plan* sub-condition marginally increased the percentage of customers vaccinated within 15-30 minutes before or after each of the popular times by 0.03 percentage points (a 3.99% average increase;  $t(9) = 1.85$ , p-value = 0.064) and significantly increased the percentage of customers vaccinated within 15-30 minutes or 15-120 minutes before or after each of the popular times (15-60 minutes before and after: a 0.08 percentage point or 3.99% average increase;  $t(9) = 3.81$ ; p-value = 0.004; 15-120 minutes before and after: a 0.10 percentage point or 2.66% average increase;  $t(9) = 2.35$ ; p-value = 0.043). However, we saw no corresponding increase in vaccinations one day before or after the day of the week of each of the ten popular times (on average, the estimated decrease in vaccinations was an insignificant 0.01 percentage points or a 0.33% decrease;  $t(9) = -0.30$ ; p-value = 0.772).

**FIGURE 4**

HEATMAP OF THE IMPACT OF PERSONALIZED SUGGESTED PLANS ON CHANGES  
IN VACCINATION RATES AT OR NEAR CUSTOMERS' LAST VACCINATION DAY OF  
THE WEEK AND TIME OF DAY



NOTE.—In the *personalized suggested plan* sub-condition, the suggested vaccination day of the week and time matched the day of the week and time of day a customer was last vaccinated, rounded to the nearest 30-minute increment. Each cell shows the difference in vaccination rates between the *personalized suggested plan* sub-condition and *standard planning prompt* condition. For each day-of-the-week/time-of-day slot, we calculated the following: (1)

the percentage of all customers in the *standard planning prompt* condition who were vaccinated in that time slot, and (2) the percentage of all customers in the *personalized suggested plan* sub-condition who were vaccinated in that time slot. We then subtracted (1) from (2) to obtain the increase/decrease in the percentage of customers vaccinated in a given time slot. These differences were then mapped into the heatmap: green cells represent increased vaccination rates, red cells represent decreased vaccination rates, and darker shades indicate larger differences.

*Heterogeneity in the Main Effect of Assignment to the Planning Prompt Offering a Suggested Plan Condition.* In additional preregistered exploratory analyses, we examined whether the main effects of our treatment (assignment to the *planning prompt offering a suggested plan* condition) varied based on consumer characteristics. After adjusting for multiple hypothesis testing using the Benjamini–Hochberg procedure (Benjamini and Hochberg 1995), assignment to the *planning prompt offering a suggested plan* condition disproportionately benefited older customers ( $\beta = 0.0001$ ,  $t(1,012,133) = 3.51$ , BH-adjusted p-value = 0.003) and, relatedly, customers with Medicare insurance ( $\beta = 0.0049$ ,  $t(1,012,131) = 3.11$ , BH-adjusted p-value = 0.028). In contrast, we find no evidence of treatment effect heterogeneity by: (1) gender, (2) receipt of a flu vaccine at our pharmacy partner in the prior flu season, (3) cumulative number of flu shots received at our pharmacy partner over the past nine years, (4) Medicaid insurance, (5) other insurance, or (6) total prescription pickups at our pharmacy partner in the previous year. See Web Appendix H for details.

## Discussion

Consistent with our theorizing, Study 1 demonstrates that planning prompts offering a suggested plan are more effective than standard planning prompts alone. Augmenting a standard planning prompt in a vaccination reminder text message by offering a suggested plan increased (1) the rate at which consumers scheduled flu vaccination appointments using links supplied to them by 0.22 percentage points (a 31% increase) and (2) the rate at which they received vaccinations at our pharmacy partner by 0.28 percentage points (a 2.45% increase). Whether the suggested plan was based on popular vaccination times or was personalized based on a customer's previous vaccination time did not affect the effectiveness of a suggested plan. Finally, our results also suggest that consumers used our suggested plans as a reference point. Customers who received a suggested plan were more likely to obtain vaccinations at or adjacent to the specified date and time, compared to those who received a planning prompt without any suggested plan.

## **STUDY 2: HOW DO PLANNING PROMPTS OFFERING A SUGGESTED PLAN COMPARE AGAINST A CURRENT BEST PRACTICE APPROACH TO NUDGING VACCINATION?**

In Study 2, we examine how the planning prompts offering a suggested plan that proved effective in Study 1 compare against the current best practice reminder messaging for encouraging vaccination.<sup>4</sup> To do so, we report the results of a preregistered, self-contained,

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<sup>4</sup> Many experiments testing reminders to encourage vaccination compare new interventions against a holdout control (Brody et al. 2026; Dai et al. 2021; Meeldijk et al. 2025; Milkman et al.

experiment embedded within the infrastructure of a larger megastudy, which tested eight different treatments designed by different teams to encourage COVID-19 booster vaccination. As noted previously, Milkman et al. (2024) reported a metanalysis of the results of all nine arms in the aforementioned megastudy, emphasizing that offering customers free round-trip Lyft rides to their vaccination appointments generated no additional vaccinations above sending mere reminder messages. Here we provide an original report of the preregistered analyses described in a self-contained experiment within the Milkman et al. (2024) megastudy about free rides.<sup>5</sup> Our experiment randomly assigned participants to two specific, preregistered message conditions. We also present original analyses of anchoring and adjustment behavior from the two arms in our experiment. Our study design and analyses were preregistered on the Open Science Framework

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2021, 2022). In contrast, we pre-registered benchmarking our reminder message's effectiveness against a reminder message that previous research identified as the best-performing vaccination reminder (Dai et al. 2021). We do this to determine whether our new reminder can outperform the strongest existing, widely-adopted reminder benchmark.

<sup>5</sup> Megastudies are designed to aggregate and compare multiple interventions within a shared infrastructure, much like a meta-analysis. Because they encompass independent experiments testing unrelated hypotheses, they are not intended to substitute for manuscripts that provide the full theoretical motivation, design logic, and reporting of pre-registered analyses in their component sub-studies (as discussed in Duckworth & Milkman, 2022). It is typical for individually pre-registered sub-studies to be reported in dedicated standalone papers that offer full theoretical and empirical accounts of their results (often alongside additional original studies) as we do here (e.g., see Kirgios et al., 2020; Buttenheim et al., 2022; Brody et al., 2026)

(OSF) (<https://bit.ly/3VLJero>). The cleaned and de-identified dataset, analysis code, log files, and table outputs for Study 2 are all posted on OSF (<https://bit.ly/4gHe8dr>).

## Participants

A total of 985,145 customers were included in the preregistered conditions of our experiment (see Web Appendix I for eligibility criteria).<sup>6</sup>

As shown in Web Appendix table K1, 59.6% of customers in our experiment were female and 40.4% were male. The average customer age was 47.3 (SD = 17.2). These customers had previously received an average of 1.77 COVID-19 vaccinations (SD = 0.83) and 0.66 total COVID-19 booster vaccinations (SD = 0.59).

## Procedure

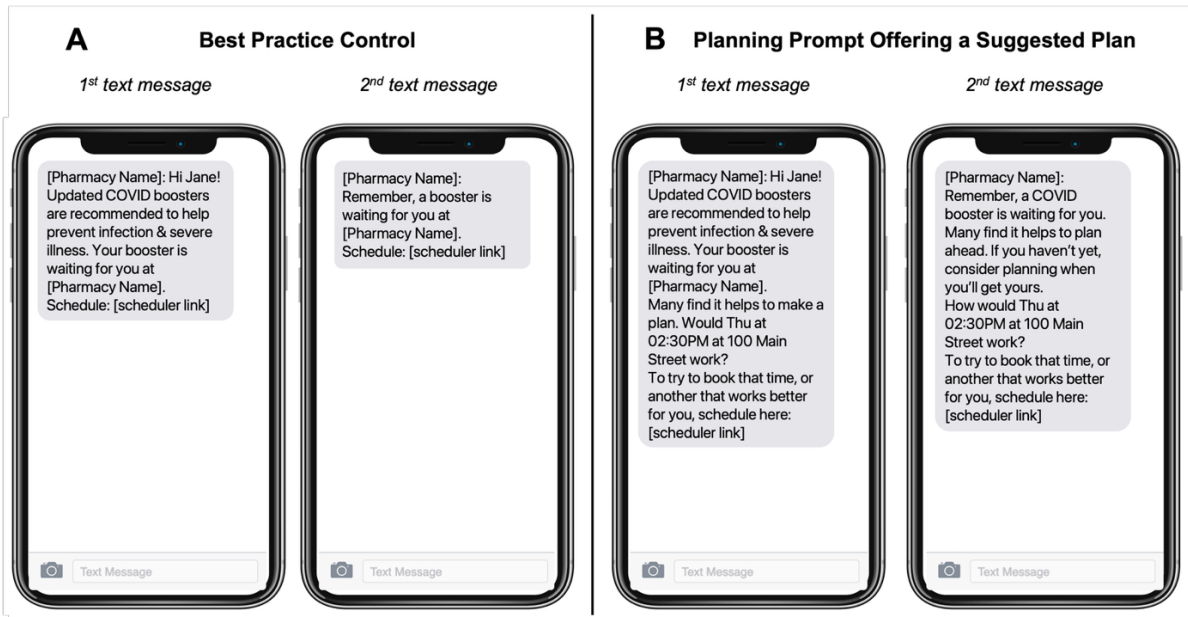
Participants in this study were the subset of megastudy participants who were randomly assigned to one of two main conditions of interest, which determined the text messages they received in the fall of 2022: a best practice control message or a planning prompt offering a suggested plan.

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<sup>6</sup> Our preregistration stated that we aimed to recruit approximately 520,000 customers—the maximum number the pharmacy expected to include given business priorities and constraints. However, our pharmacy partner was ultimately able to secure a larger sample size than initially planned, resulting in a total of 985,145 customers randomized to eligible conditions.

As shown in Figure 5, customers who were assigned to the *best practice control* condition were told that a COVID-19 vaccine was “recommended” and “waiting for [them]” and were prompted to schedule a vaccination using an online vaccine scheduler link. This best practice message—which employs psychological ownership framing and does not include a planning prompt or a suggested plan—has proved more effective than other types of reminder messages in prior experiments on encouraging flu vaccination (Bogard, Fox, and Goldstein 2021; Milkman et al. 2021, 2022) and COVID-19 vaccination (Dai et al. 2021). Customers who were assigned to the *planning prompt offering a suggested plan* condition received a message that was identical to the *best practice control* condition message except that it also included a planning prompt: “Many find it helps to make a plan. Would [day of the week] at [time of day] at [pharmacy address] work? To try to book that time or a time that works better for you, schedule here,” where the day of the week, time of day, and location suggested were selected to match when and where the customer had received their last vaccination at our pharmacy partner (i.e., the suggested plan was personalized in exactly the same way it was in Study 1). Customers in both conditions also received a second reminder message, mirroring the contents of the first reminder, which arrived seven days after the initial reminder message.

The initial text messages were sent out to customers on three dates in November 2022: (1) 328,740 messages were sent on November 3<sup>rd</sup>, (2) 327,980 messages were sent on November 5<sup>th</sup>, and (3) 328,425 messages were sent on November 8<sup>th</sup>. Reminder messages were then sent to customers seven days after the initial text message on November 10<sup>th</sup>, November 12<sup>th</sup>, and November 15<sup>th</sup>, respectively.

**FIGURE 5****TEXT MESSAGES SENT TO CUSTOMERS BY CONDITION IN STUDY 2**

NOTE.—In the *planning prompt offering a suggested plan* condition, the suggested day of the week, time of day, and pharmacy location matched the day of the week, time of day, and pharmacy location of the customer's most recent vaccination at our pharmacy partner.

### Dependent Variables and Analysis Strategy

As shown in table 4, we defined three categories of dependent variables. We analyzed whether random assignment to the *planning prompt offering a suggested plan condition* affected each dependent variable as follows:

**TABLE 4****DEFINITION OF DEPENDENT VARIABLES IN STUDY 2**

| <b>Dependent Variable</b>                                                  | <b>Definition of Dependent Variable</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Vaccinations Received</i>                                               | Our preregistered primary dependent variable was whether a participant received a bivalent COVID-19 booster at our pharmacy partner within 30 days of the send date of their initial reminder. Two secondary dependent variables measuring vaccination receipt were also preregistered: (1) receipt of a bivalent COVID-19 booster within 60 days and (2) receipt of a bivalent COVID-19 booster within 90 days of the send date of a customer's initial reminder message.                                                                                                                                                                                        |
| <i>Vaccinations Received by Following Our Suggested Plan</i>               | Every suggested plan received by customers recommended the day of the week of the customer's most recent prior vaccination. Accordingly, we preregistered analyzing whether customers received a bivalent COVID-19 booster at our pharmacy partner within 30 days of their initial reminder on the day of week of their last vaccination. Because the dataset provided by our pharmacy partner for this experiment did not include information on the time of day or location of customer vaccinations, we were only able to test whether customers received a vaccination on the day of the week of their last vaccination (consistent with our suggested plan). |
| <i>Vaccinations Received by Deviating Slightly from Our Suggested Plan</i> | In analyses that were not preregistered, we analyze whether customers received a bivalent COVID-19 booster at our partner pharmacy, within 30 days of the send date of their initial reminder, on the day before or after the day of the week of their last vaccination.                                                                                                                                                                                                                                                                                                                                                                                          |

*Vaccinations Received.* Following Milkman et al. (2024), we deviate from our preregistration to address an unexpected issue with the vaccination data provided to us, employing a more conservative analysis strategy than originally planned (an intent-to-treat analysis; see Web Appendix J for details). We estimate an ordinary least squares (OLS) regression with HC1 robust standard errors and without control variables to predict whether a

given customer received a bivalent COVID-19 booster within 30 days of the first text message sent to them. The sole predictor variable is an indicator for assignment to the *planning prompt offering a suggested plan* condition.

*Vaccinations Received by Following Our Suggested Plan.* In preregistered exploratory analyses not reported in Milkman et al. (2024), we conducted a two-sample proportion test to assess whether customers who were randomly assigned to the *planning prompt offering a suggested plan* condition were more likely than those assigned to the *standard planning prompt condition* to get vaccinated on the same day of the week as their most recent prior vaccination.

*Vaccinations Received by Deviating Slightly from Our Suggested Plan.* In analyses that were not preregistered and not reported in Milkman et al. (2024), we conducted a two-sample proportion test to assess whether customers who were randomly assigned to the *planning prompt offering a suggested plan* condition were more likely than those assigned to the *standard planning prompt condition* to get vaccinated on the day before or after the day of the week of their most recent prior vaccination.

## Results

*Balance Checks.* As shown in Web Appendix table K1, customers in the two study conditions were balanced on age, gender, insurance type, total prior COVID-19 vaccinations, and total prescription pickups at our pharmacy partner in the previous year (all p-values > 0.144). However, we observed an imbalance in the number of total prior COVID-19 boosters as of

October 2022 (prior to the launch date of the study). To address this unexpected imbalance, we report all regression results as preregistered as well as with an additional control for total prior COVID-19 boosters as of October 2022; all results are robust to this adjustment and remain substantively unchanged.

*Vaccinations Received.* As shown in Web Appendix table L1, model 1, which presents our primary preregistered regression analysis, assignment to the *planning prompt offering a suggested plan* condition increased 30-day vaccine uptake by a regression-estimated 0.20 percentage points compared to assignment to the *best-practice reminder* condition (a 3.26% increase from 6.09%;  $t(985,143) = 4.09$ ,  $p < 0.001$ ). Our results are robust to including a single control variable for a customer's number of prior COVID-19 booster shots (see Web Appendix table L1, model 2), to including all control variables from our original preregistered model specification (see Web Appendix table L1, model 3), and additional preregistered robustness checks on primary and secondary dependent variables (see Web Appendix table L2, M1 and M2). For full reporting of all preregistered heterogeneity results, see Web Appendix N.

*Vaccinations Received by Following Our Suggested Plan.* Consistent with evidence for anchoring, more customers elected to receive a vaccination on the same day of the week as their previous vaccination in the *planning prompt offering a suggested plan* condition (1.38%) than in the *best practice control* condition in which no suggested plan was provided (1.20%), a 0.18 percentage point increase (or a 15.26% increase;  $\chi^2(1) = 64.64$ ,  $p < 0.001$ ).

*Vaccinations Received by Deviating Slightly from Our Suggested Plan.* Consistent with anchoring and adjustment, assignment to the *planning prompt offering a suggested plan* condition marginally increased the likelihood that customers received their vaccination one day before or after the suggested day of the week by 0.05 percentage points (a 2.88% increase;  $\chi^2(1) = 3.67$ ;  $p\text{-value} = 0.055$ ).

## Discussion

Study 2 extends the results of Study 1 by showing that planning prompts offering a suggested plan are not only better than standard planning prompts at encouraging vaccination (as shown in Study 1) – they are more effective than best practice reminders for encouraging vaccination. As in Study 1, we also find some support for the theory that consumers may treat suggested plans as an anchor or reference point. Customers who were assigned to receive a planning prompt with a suggested plan were more likely to get vaccinated on the day of the week of their last vaccination (recommended by our suggested plans), and marginally more likely to receive vaccinations on the day adjacent to the day of the week of their last vaccination.

## GENERAL DISCUSSION

In two field experiments (total  $N = 1,997,288$ ), we demonstrate that planning prompts offering a suggested plan are more effective than standard planning prompts or best practice vaccination messages. Specifically, we show that planning prompts offering a suggested plan increase influenza flu vaccination rates by 2.45% relative to standard planning prompts (Study 1)

and increase COVID-19 booster uptake by 3.26% over best-practice reminders (Study 2). Our results suggest that providing a suggestion facilitates the planning process and makes it easier to follow through on an intention. Including a suggested plan increased the likelihood that consumers formed a plan by booking an appointment through our text scheduler. Moreover, suggested plans seemed to provide consumers with a reference point that served as an anchor, influencing their eventual vaccination time and location: this implies that the inclusion of a suggestion provided a heuristic or mental shortcut they used when deciding when and where to get vaccinated. This suggests that simplifying the planning process by suggesting a plan is a benefit that can outweigh the drawback of discouraging consumers from self-generating plans (i.e., potentially reducing autonomy, memorability, and commitment). Lastly, we find that suggested plans are just as effective in our studies when they are based on group norms rather than “personalized” based on a specific consumer’s past behavior.

Our findings offer new insights about the best way to generate plans, contributing to a large literature on the psychology of planning and follow-through (Gollwitzer and Sheeran 2025; Oettingen 2015; Rogers et al. 2015). Prior research has theorized that providing examples of plans might hinder a consumer’s ability to identify optimal cues to help them follow through on their intentions (Hagger and Luszczynska 2014). This notion is supported by a recent meta-analysis showing stronger benefits from planning when consumers specify their own cues to trigger action when generating implementation intentions or plans (Sheeran et al. 2025). However, our random assignment of consumers to conditions in an externally valid setting shows that providing consumers with a suggested plan can improve follow-through, at least in this study’s context. One driver of this difference may be the planning context itself: prior planning studies typically required participants to formulate a plan, whereas our naturalistic context could

only encourage it. In such a voluntary setting, the benefit of making planning easier with a suggested plan may therefore outweigh the benefits of self-generating one.

We also contribute to the literature on how suggestions change consumer behavior. Prior research on offering consumers suggestions has predominantly examined how they affect consumers' decisions on numerical quantities, including charitable donations (Fraser et al. 1988; Goswami and Urminsky 2016; Moon and VanEpps 2023), tipping decisions (Alexander et al. 2021; Haggag and Paci 2014), and product ratings (Yadav 1994). Our studies extend this research by examining how suggestions affect planning. Our results show that a suggested plan for when and where to take an encouraged action increases the likelihood that consumers will take that action on or close to the suggested day of the week, the suggested time, and the suggested location. This finding aligns with the possibility that offering a suggestion creates a kind of reference point (Tversky and Kahneman 1974), which we know from past research facilitates recall (Strack and Mussweiler 1997), triggers confirmatory hypothesis testing (Chapman and Johnson 1999), and can serve as an anchor from which consumers may adjust (Epley and Gilovich 2006).

Finally, we contribute to the growing literature on personalization by showing that personalization using prior consumer behavioral data does not make suggested plans any more potent.

### Limitations and Future Directions

Our study has a number of limitations. First, both of our experiments targeted vaccine adherence. Future research should explore whether planning prompts offering a suggested plan

can improve consumer follow-through on a more diverse set of outcomes, such as health screenings (Lieberman et al. 2021), healthy food choices (John, Donnelly, and Roberto 2017; VanEpps, Downs, and Loewenstein 2016), environmentally sustainable consumption (Schwartz, Milfont, and Hilton 2019) and timely debt repayment (Mazar, Mochon, and Ariely 2018).

Second, our study tested the value of suggested plans generated based on rich data about prior consumers' behavior. It is possible that providing lower-quality plans (e.g., randomly generated appointment times and locations) could attenuate our effect, since recommending inferior choices could potentially generate distrust (Bogard et al. 2024) or provoke reactance (Fitzsimons and Lehmann 2004). Third, because offering a suggested plan lengthened our reminder messages, consumers in our treatment conditions received slightly longer reminders (see figures 2 and 5) than those in our control conditions. Prior research has not found a correlation between the length of concurrent reminder messages and vaccination reminder effectiveness (Milkman et al. 2022), and theory suggests simpler messages are generally more effective (Sunstein 2013).

However, it is a limitation of our study that we could not hold reminder message length constant across conditions. Fourth, because our pharmacy partner routinely sends appointment reminders to consumers who have already scheduled a visit, it is possible that the theorized benefit of improved recall for standard planning prompts is attenuated in our context, where reminders are already salient and timely—a feature common to many appointment-based settings. Finally, our intervention produced relatively small improvements in vaccination rates over and above other effective nudges currently in use (e.g., a standard planning prompt or a best-practice vaccination reminder message). As a result, a planning prompt offering a suggested plan may only be worth deploying instead of a standard planning prompt or best-practice vaccination reminder when the marginal cost of offering consumers high-quality suggested plans is low, or in settings where it

leads to significantly larger benefits than those observed in our studies such that the benefits outweigh the costs.

Future research could also build upon our theoretical contributions by exploring additional psychological mechanisms through which suggested plans operate. For example, it could be that presenting a suggested plan may give consumers the illusion that they have already made goal progress (a plan already exists, after all). This perception can increase both their commitment to their goal and their likelihood of following through by enhancing the perceived ease of execution. Prior research on goal pursuit suggests that the subjective impression of initial goal progress increases goal commitment and completion by lowering the perceived difficulty of achieving the goal (Nunes and Drèze 2006; Zhang and Huang 2010).

Another profitable avenue for future research entails mapping the boundary conditions of suggested plans. When making choices, consumers may have different goals: sometimes they prioritize minimizing cognitive effort, while at other times they prioritize maximizing decision quality (i.e., selecting the choice with the highest utility; Bettman et al. 1998). Planning prompts offering a suggested plan may be most effective in contexts where consumers are seeking to minimize their cognitive effort and may be less valuable in settings where consumers are focused on optimizing their decisions' quality and therefore willing to invest meaningful time and resources to construct an ideal plan (e.g., planning the day, time, and location for a wedding).

The effectiveness of a suggested plan may also depend on decision complexity and the strength of consumer preferences—factors that generally moderate the effectiveness of all nudges (Thaler and Sunstein 2009). If a suggested plan simplifies decision-making, it may be most effective when consumers face complex decisions, such as decisions where no clear dominant option exists (Dhar 1997). A suggested plan may also be more impactful when

consumers cannot articulate their ideal choice (Chernev 2003) or when a decision-maker's preferences are unclear or unknown (Johnson and Goldstein 2013). In such cases, accepting a suggestion allows consumers to avoid the cognitive effort required to construct or clarify their preferences (Bettman et al. 1998). Future research exploring these predictions would be valuable.

### Practical Implications for Marketers, Managers, and Policy Makers

Marketers, managers, and policy makers frequently struggle to motivate consumers to act on their intentions (Sheeran 2002). Our findings suggest an actionable strategy: use planning prompts with suggested plans to close intention-behavior gaps. A planning prompt offering a suggested plan could (1) facilitate appointment scheduling for sales leads (e.g., by suggesting a date for a financial consultation), (2) help consumers follow-through on dream vacations (e.g., by suggesting a date for a weekend trip), or even (3) encourage timely product upgrades (e.g., by suggesting a time for a smartphone upgrade before a trade-in offer expires). In the context of vaccination, we show that planning prompts offering a suggested plan are useful at scale and can influence a diverse consumer population. Because these planning prompts do not increase the marginal cost of sending a reminder, they are also extremely cost-effective to implement in contexts where reminders are already planned and personalization is straightforward.

Suggested plans can also substitute for other effective but costly behavioral interventions. For example, providing consumers with default appointments is an extremely effective way to promote vaccination, but massively increases appointment no-shows (e.g., by 71 percentage points in Chapman et al., 2016). In contexts where reserving capacity is expensive (e.g., physician visits or other scheduled services), the opt-in nature of suggested plans offers an

advantage: consumers are more likely to follow-through conditional on booking an appointment, allowing firms to reserve supply to meet demand more predictably.

Our findings also indicate that extensive personalization is not required for marketers and policy-makers to achieve benefits from planning prompts offering a suggested plan. Remarkably, suggesting a set of popular times and days for vaccination is just as effective as providing a personalized suggested plan based on a consumer's past behavior. This distinction matters in practice: personalized suggestions require substantial infrastructure—including data retention, algorithmic systems, and ongoing maintenance—for delivery at scale, whereas popular suggestions do not, making them particularly attractive for resource-constrained organizations. Popular suggestions also offer greater flexibility, as they provide a larger set of potential recommendations per person and can also be used in communications with consumers who have never engaged in the target behavior before and so could not be offered a suggested plan based on their prior activity.

## Summary

In conclusion, we present evidence from two large field experiments demonstrating that planning prompts offering a suggested plan increase consumer follow-through. Planning prompts offering a suggested plan outperform both standard, open-ended planning prompts and best-practice reminder messages as a means of encouraging vaccination, albeit by small margins. Moreover, planning prompts offering a suggested plan can replace other reminders at near zero marginal cost, generating reliable benefits that persist for months post-intervention. Overall, our

field experiments document the value of a new nudge that we anticipate marketers can use to close the intention-behavior gap in a wide range of settings.

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## DATA COLLECTION STATEMENT

Study 1 was conducted in partnership with a national U.S. pharmacy chain in September 2023. Study 2 was conducted in partnership with a national U.S. pharmacy chain in November 2022. The data contain proprietary patient information. For each study, the authors signed a non-disclosure agreement (NDA) with a pharmacy partner that prevents them from naming the partner without their permission. Therefore, the pharmacy partner for each study cannot be named unless they approve disclosure. The pharmacy partner for each study collected the data and provided de-identified data in accordance with the Safe Harbor method pursuant to 45 C.F.R. § 164.514(b)(2). All data were transferred and stored in secure, access-restricted project directories on password-protected servers at the University of Pennsylvania's Leonard Davis Institute of Health Economics Health Services Research Data Center. For Study 1, Robert Kuan cleaned and analyzed the data; Youngwoo Jung and Kevin V. Zhu jointly replicated the analyses. For Study 2, we use data from two conditions in Milkman et al. (2024), which has already been made publicly available on the Open Science Framework (<https://bit.ly/3MhRHgm>), along with additional variables provided by our pharmacy partner at the time that were not included in the megastudy analysis. Robert Kuan again cleaned and analyzed the data, with replication by Youngwoo Jung and Kevin V. Zhu. Due to the sensitive nature of this data, the raw dataset and the data cleaning code for both studies cannot be shared; however, fully anonymized, de-identified data—including each participant's intervention condition, launch date, vaccination outcomes during our follow-up period, and other relevant variables—are available on the Open Science Framework (Study 1: <https://bit.ly/3KmXZhl>; Study 2: <https://bit.ly/4gHe8dr>). Further, should any researcher have an interest in replicating our data cleaning code, we would work in

good faith to obtain permission from our U.S. pharmacy chain partner to add that researcher to our NDA and grant them access to the original data on the secure server for replication purposes.